Supplementary Appendix

Supplement to: Schmid P, Cortes J, Dent R, et al. Overall survival with pembrolizumab in early-stage triple-negative breast cancer. N Engl J Med 2024;391:1981-91. DOI: 10.1056/NEJMoa2409932

This appendix has been provided by the authors to give readers additional information about the work.

Supplementary Appendix

For

Overall Survival with Pembrolizumab in Early-Stage Triple-Negative Breast Cancer

Authors: Schmid et al.

TABLE OF CONTENTS **Page**

List of Investigators 2
Steering Committee 9
Supplemental Methods 10
Supplemental Results..... 10
Figure S1. Trial Enrollment and Outcomes of the Patients at the Seventh Interim Analysis 11
Figure S2. Additional Subgroup Analyses of Overall Survival..... 12
Figure S3. Additional Subgroup Analyses of Event-Free Survival 13
Figure S4. Overall Survival by Pathological Complete Response (ypT0/Tis ypN0)..... 14
Table S1. Demographic and Disease Characteristics at Baseline 15
Table S2. Representativeness of Patient Population..... 17
Table S3. Summary of Drug Exposure in the Combined Phases.....19

List of Principal Investigators.

Country	Site Name	Principal Investigator
Australia	Cabrini Health	Antill, Yoland
	Westmead Hospital	Dinh, Phuong/Hui, Rina
	Royal Brisbane and Women's Hospital	Inglis, Po-ling
	Royal North Shore Hospital	Menzies, Alexander M./Baron-Hay, Sally
	Royal Adelaide Hospital	Murray, Nick
	Frankston Hospital	Thomson, Jacqui
	St. John of God Subiaco Hospital	Tsoi, Daphne
Brazil	Uniao Brasileira de Educacao e Assistencia Hospital Sao Lucas da Pucrs	Barrios, Carlos Henrique E.
	Hospital de Base de Sao Jose de Rio Preto	Cardoso Guedes, João Daniel/Girotto, Gustavo Colagiovanni
	UOPECCAN - Uniao Oeste Paranaense de Estudos e Combate ao Cancer	da Cunha, Ademar Dantas Jr.
	Universidade de Caxias do Sul	de Cassia Costamilan, Rita
	Instituto do Cancer do Ceara	de Figueiredo Chaves, Fabio
	Hospital Araujo Jorge Associacao de Combate ao Cancer de Goias	de Freitas, Ruffo Jr.
	Instituto do Cancer do Estado do Sao Paulo – ICESP	Linck, Rudinei Diogo Marques
	Hospital Nossa Senhora da Conceicao	Pedrini, Jose Luiz
	Hospital Erasto Gaertner	Skare, Nils Gunnar
Canada	Jewish General Hospital	Boileau, Jean-Francois
	Princess Margaret Cancer Centre	Cescon, David
	Centre Hospitalier de l Universite de Montreal – CHUM	Hassan, Saima/Robidoux, Andre
	CIUSSS de l'Estrie-CHUS	Pavic, Michel
	CHU de Quebec Universite Laval - Hopital du Saint-Sacrement	Poirier, Brigitte/Provencher, Louise
	The Ottawa Hospital - Cancer Care	Song, Xinni
	Tom Baker Cancer Centre	Webster, Marc/Verma, Sunil
Colombia	Hemato Oncologos S.A.	Avendano Rojas, Ana Cristina
	Instituto Nacional de Cancerologia (Bogota, Colombia)	Bruges, Ricardo
	Instituto De Cancerologia S.A.	Gonzalez, Diego Mauricio/Montoya Restrepo, Maria Elvira
	Oncomedica S.A.	Gonzalez Fernandez, Manuel Enrique
	Hospital Universitario San Ignacio	Pinilla, Paola Andrea/ Sanchez Castillo, Jesus Oswaldo/Rojas, Luis Leonardo
	Oncologos del Occidente S.A.	Rojas-Uribe, Gustova Adolfo
France	Clinique Victor Hugo	Bourgeois, Hugues

	CHU de la Miletrie Poitiers	Chieze, Stephanie
	Hopital prive du Confluent	Chocteau-bouju, Dorothee
	Institut Claudius Regaud IUCT Oncopole	Dalenc, Florence
	Hopital Diaconesses Croix Saint Simon	Delbaldo, Catherine
	Polyclinique Bordeaux Nord Aquitaine	Dohollou, Nadine
	Hopital Saint Louis	Giacchetti, Sylvie
	Centre Francois Baclesse	Levy, Christelle
	CHU Jean Minjoz	Mansi, Laura/Chaigneau, Loic
	Centre Jean Perrin	Mouret-Reynier, Marie-Ange
	Institut Curie	Sablin, Marie-Paule/Loirat, Delphine
Germany	Caritasklinik St. Theresia	Deryal, Mustafa
	Universitaetsklinikum Erlangen	Fasching, Peter A.
	Universitaets-Frauenklinik Tuebingen	Grischke, Eva-Maria/Engler, Tobias/Hartkopf, Andreas
	Klinikum der Universit. Muenchen	Harbeck, Nadia
	Gynaekologisches Zentrum	Kurbacher, Christian Martin
	Universitaetsklinikum Hamburg-Eppendorf	Mueller, Volkmar/Witzel, Isabell
	Kliniken Essen Mitte	Reinisch, Mattea
	Universitaetsklinik und Poliklinik Halle/Saale	Thomssen, Christoph
	HELIOS Klinikum Berlin-Buch	Untch, Michael
Ireland	Bon Secours Hospital	Murphy, Conleth
	St Vincent's University Hospital	Walshe, Janice
	Yitzhak Shamir Medical Center	Merose, Rotem/Rasco, Adi/Katz, Daniella
	Sheba Medical Center	Nili-Gal-Yam, Einav/ Kaufman, Bella/Paluch-Shimon, Shani
	Hadassah Ein Karem – Sharett Institute of Oncology Kiryat Hadassah	Peretz-Yablonski, Tamar
	Rabin-Medical Center	Stemmer, Salomon
	Oncology Institute	Tokar, Margarita/Geffen, David/
	Tel Aviv Sourasky Medical Center	Wolf, Ido/Evron, Ella/ Ryvo, Larisa
Italy	Ospedale San Luca	Baldini, Editta
	Ospedale Civile di Macerata	Battelli, Nicola/Beneditti, Giovanni
	Istituto Europeo di Oncologia	Colleoni, Marco
	Istituto Nazionale Tumori IRCCS Fondazione Pascale	De Laurentiis, Michelino
	Istituto Scientifico Romagnolo per Studio e Cura Tumori IRST	di Giorgi, Ugo/Rocco, Andrea
	Azienda Socio Sanitaria Territoriale degli Spedali Civili di Brescia	Pedersini, Rebecca
Japan	Chiba Cancer Center	Hayama, Shoko/Yamamoto, Naohito

	Hiroshima City Hiroshima Citizens Hospital	Itoh, Mitsuya/Ohtani, Shoichiro
	Kindai University Hospital	Komoike, Yoshifumi
	Aichi Cancer Center Hospital	Kotani, Haruru/Iwata, Hiroji/Hattori, Masaya
	National Hospital Organization Hokkaido Cancer Center	Maeda, Hideki/Takahashi, Masato
	Hyogo College of Medicine Hospital	Miyoshi, Yasuo
	National Cancer Center Hospital East	Mukai, Hirofumi
	Saitama Cancer Center	Nagai, Shigenori/Inoue, Kenichi
	Tokai University Hospital	Niikura, Naoki/Tokuda, Yutaka
	Saitama Medical University International Medical Center	Osaki, Akihiko/Saeki, Toshiaki
	Social Medical Corporation Hakuaikai Sagara Hospital	Sagara, Yasuaki
	The Cancer Institute Hospital of JFCR	Takano, Toshimi/Ohno, Shinji
	Shizuoka Cancer Center Hospital and Research Institute	Tadokoro, Yukiko/Watanabe, Junichiro
	Toranomon Hospital	Tanabe, Yuko/Takano, Toshimi
	St. Marianna University School of Medicine Hospital	Tsugawa, Koichiro
	Kumamoto University Hospital	Yamamoto, Yutaka
	National Hospital Organization - Osaka National Hospital - Institute for Clinical Research	Yasojima, Hiroyuki/Masuda, Norikazu
	National Cancer Center Hospital	Yonemori, Kan/Shimizu, Chikako/Tamura, Kenji/
	St. Luke's International Hospital	Yoshida, Atsushi/Hayashi, Naoki
Poland	Mazowiecki Szpital Onkologiczny	Bauer-Kosinska, Barbara
	Centrum Onkologii im. Prof. Franciszka Lukaszczyka	Chmielowska, Ewa
	Centrum Onkologii Ziemi Lubelskiej im. sw. Jana z Dukli	Chrzanowska-Kapica, Agata
	Szpital Morski im. PCK. Szpitale Pomorskie Sp. Z o.o.	Danielewicz, Iwona/Lesniewski Kmak, Krzysztof
	Narodowy Instytut Onkologii im. Marii Sklodowskiej-Curie - Oddzial w Krakowie	Drosik Kwasniewska, Anna Patrycja/Grela-Wojewoda, Aleksandra
	Dolnoslaskie Centrum Onkologii	Lacko, Aleksandra
	Narodowy Instytut Onkologii - Oddzial w Gliwicach	Tarnawski, Rafal
	Centrum Onkologii im. Prof. Franciszka Lukaszczyka	Zurawski, Bogdan

Portugal	Champalimaud Clinical Centre/Champalimaud Foundation	Cardoso, Fatima
	Hospital de Santa Maria, E.P.E.	Castro Abreu Ribeiro, Leonor/ Dionísio, Maria Rita/ Sousa, Rita Teixeira
	Instituto Português de Oncologia do Porto Francisco Gentil, EPE	Ferreira, Marta
Republic of Korea	Asan Medical Center	Ahn, Jin-Hee
	Seoul National University Hospital	Im, Seock-Ah
	Severance Hospital Yonsei University Health System	Kim, Min Hwan/Kim, Gun Min
	Samsung Medical Center, Sungkyunkwan University School of Medicine	Park, Yeon Hee
Russia	Arkhangelsk Clinical Oncological Dispensary	Chapko, Yana S./Lebedeva, Lyudmila N./Nechaeva, Marina N.
	Chelyabinsk Regional Clinical Oncological Dispensary	Fadeeva, Natalia Vladmirovna
	Russian Oncological Research Center n.a. N.N. Blokhin of MoH	Ganshina, Inna/Lichinitser, Mikhail Romanovich/
	Scientific Research Oncology Institute n.a. N.N. Petrov	Krivorotko, Petr Vladimirovich
	Republican Clinical Oncology Dispensary of Republic of Bashkortostan	Lipatov, Oleg Nikolaevich
	Republican Clinical Oncology Dispensary of Tatarstan MoH	Mukhametshina, Guzel Zinnurovna
	GBU RO Regional Clinical Oncological Dispensary	Shomova, Marina V.
Singapore	National Cancer Centre Singapore	Dent, Rebecca
Spain	Hospital del Mar	Albanell Mestres, Joan
	Hospital Clinico Univ de Valencia	Bermejo de las Heras, Begona
	Hospital Universitario Reina Sofia	de la Haba Rodriguez, Juan
	Hospital Universitario Ramon y Cajal	Gión Cortes, Maria/Holgado Martin, Esther
	Hospital Quiron de Madrid	Gonzalez Cortijo, Lucia
	Complejo Hospitalario Universitario de Santiago	Lopez Lopez, Rafael
	Hospital Universitario Virgen del Rocio	Salvador Bofill, Javier
	Instituto Oncologico Baselga, Hospital Quiron	Soberino Garcia, Jesus/Perez Garcia, Jose Manuel
	ICO L Hospitalet	Villanueva Vazquez, Rafael/Morilla Ruiz, Idoia
	Hospital General Universitari Vall d Hebron	Zamora Adelantado, Esther/Munoz Couselo, Eva
Sweden	Linkopings Universitetssjukhus Onkologiska kliniken	Albu-Kareem, Ahmed/Hallbeck, Anna-Lotta
	Norrlands Universitetssjukhus Universitetssjukhuset Cancercentrum	Andersson, Anne

	Karolinska Universitetssjukhuset Solna	Foukakis, Theodoros
	Akademiska Sjukhuset Onkologiska kliniken	Lindman, Henrik
Taiwan	MacKay Memorial Hospital	Chang, Yuan-Ching
	Chang Gung Medical Foundation, Linkou Branch	Chen, Shin-Cheh
	Koo Foundation Sun Yat-Sen Cancer Center	Chung, Chi-Feng/Liu, Mei-Ching
	National Taiwan University Hospital	Huang, Chiun-Sheng
	National Cheng Kung University Hospital	Lee, Kuo-Ting
	Taipei Veterans General Hospital	Tseng, Ling-Ming
Turkiye	Adana Acıbadem Hospital	Abali, Huseyin
	Hacettepe Universitesi Tip Fakultesi Hastanesi	Aksoy, Sercan
	I.E.U. Medical Point Hastanesi	Arslan, Cagatay
	Acibadem Altunizade Hastanesi	Basaran, Gul
	Ege University Medical Faculty Tulay Aktas Oncology Hospital	Gokmen, Erhan/Uslu, Ruchan F.
	Dr Abdurrahman Yurtaslan Oncology Training and Research Hospital	Karacin, Cengiz/Oksuzoglu, Berna
	Memorial Sisli Hastanesi	Keskin, Serkan
	Amerikan Hospital Medical	Molinas Mandel, Nil
	Ozel Medicana International Ankara Hastanesi	Ozdemir, Evren
	Antalya Memorial Hospital	Ozdogan, Mustafa
	İstanbul University Cerrahpaşa Medical Faculty	Ozguroglu, Mustafa
	Baskent Unıversity Adana Kısla Hospital	Ozyilkan, Ozgur
	Trakya University Medical Faculty Balkan Oncology Hospital	Topaloglu, Sernaz/Cicin, Irfan
	Ondokuz Mayis Universitesi	Yilmaz, Bahiddin/Yucel, Idris
United Kingdom	Nottingham University Hospitals NHS Trust	Chan, Steve
	The James Cook University Hospital	Graham, Janine Elizabeth
	Maidstone Hospital	Harper-Wynne, Catherine
	St George’s Hospital	Kyle, Fiona
	Colchester General Hospital	Mukesh, MB
	Barts Cancer Institute, Charterhouse Square, Queen Mary University of London	Schmid, Peter
	Royal Cornwall Hospitals NHS Trust	Wheatley, Duncan
United States	Henry Ford Hospital	Ali, Haythem
	Northwest Cancer Specialists, P.C.	Andersen, Jay C.
	New England Cancer Specialists	Battelli, Chiara
	Magee Women’s Hospital	Brufsky, Adam//Emens, Leisha Ann/Jankowitz, Rachel
	Texas Oncology-Memorial City	Cairo, Michelina/Holmes, Frankie Ann
	Houston Methodist Cancer Center	Chang, Jenny
	Medical Oncology Associates (Summit Cancer Centers)	Chaudhry, Arvind

	Peninsula Cancer Institute, LLC	Chevinsky, Aaron/MacLaughlin, William
	North Shore Oncology Hematology Associates	Chung, Elizabeth K./Weyburn, Thomas D.
	Virginia Oncology Associates	Danso, Michael A.
	University of Colorado Cancer Center	Diamond, Jennifer Robinson
	University of Virginia Comprehensive Cancer Center	Dillon, Patrick
	Rhode Island Hospital	Fenton, Mary Anne/Lopresti, Mary
	Scottsdale Healthcare Hospitals D/B/A HonorHealth	Gordon, Michael S./Sachdev, Jasgit
	Moncrief Cancer Institute	Haley, Barbara B.
	Parkland Health and Hospital System	Haley, Barbara B.
	Simmons Cancer Center	Haley, Barbara B.
	UT Southwestern Medical Center	Haley, Barbara B.
	Moncrief Cancer Institute	Haley, Barbara B.
	Broome Oncology, LLC	Harris, Ronald P.
	Texas Oncology-San Antonio Northeast	Harroff, Allyson/Wilks, Sharon T.
	Bon Secours St. Francis Medical Center-Oncology Research	Irvin, William J.
	Pacific Cancer Care	Koontz, Michael Zachary/Stampleman, Laura
	Goshen Center for Center Care	Kio, Ebenezer/Bruetman, Daniel G.
	Minnesota Oncology Hematology, P.A.	Larson, Timothy G./Tsai, Michaela
	University of Iowa Hospital and Clinics	Leone, Jose/Phadke, Sneha
	YVMH dba Virginia Mason Memorial/North Star Lodge Cancer Center	Mannem, Siva/Obulareddy, Sri/Jones, Vicky E.
	TOI Clinical Research	Marathe, Omkar/Thara, Eddie
	TriHealth Cancer Institute-Good Samaritan Hospital	Mehta, Apurva
	Helen F. Graham Cancer Center & Research Institute	Misleh, Jamal/Guarino, Michael J.
	Cedars Sinai Medical Center	Mita, Monica M./Basho, Reva/McArthur, Heather
	University of Chicago Medical Center	Nanda, Rita
	Orchard Healthcare Research Inc.	Oliff, Ira A.
	Texas Oncology - Baylor Charles A. Sammons Cancer Center	OShaughnessy, Joyce A.
	Providence Portland Medical Center	Page, David
	Texas Oncology-Austin Central	Patt, Debra A.
	Yale University School of Medicine	Pusztai, Lajos
	Texas Oncology-Northeast Texas	Richards, Donald A.
	Nyack Hospital Infusion Center	Rybalova, Irina

	NorthShore University Health System - Evanston Hospital	Saha, Poornima
	Seattle Cancer Care Alliance	Specht, Jennifer
	Virginia Cancer Specialists, PC	Spira, Alexander I./Denduluri, Neelima
	Texas Oncology - Plano East	Stokoe, Christopher T.F.
	Arizona Oncology Associates P.C.- HOPE	Swart, Rachel
	Rutgers Cancer Institute of New Jersey	Toppmeyer, Deborah Lynn
	University of Miami - Sylvester Comprehensive Cancer Center - Kendall satellite	Valdes, Frances
	The West Clinic, P.C.	Vidal, Gregory
	Oncology/Hematology Care Clinical Trials, LLC	Ward, Patrick
	Kadlec Clinic Hematology and Oncology	Zhuo, Ying

Steering Committee.

Peter Schmid: Barts Cancer Institute, Queen Mary University London, London, UK

Javier Cortes: International Breast Cancer Center, Quironsalud Group; Vall d'Hebron Institute of Oncology (VHIO), Madrid & Barcelona, Spain and Universidad Europea de Madrid, Faculty of Biomedical and Health Sciences, Department of Medicine, Madrid, Spain

Carsten Denkert: Institute of Pathology, Philipps-University Marburg and University Hospital Marburg (UKGM), Marburg, Germany

Rebecca Dent: National Cancer Center Singapore, Duke-NUS Medical School, Singapore

Sherko Kuemmel: Breast Unit, Kliniken Essen-Mitte, Essen, Germany and Charité – Universitätsmedizin Berlin, Department of Gynecology with Breast Center, Berlin, Germany

Heather McArthur: University of Texas Southwestern Medical Center, Dallas, TX, USA

Lajos Pusztai: Yale School of Medicine, Yale Cancer Center, New Haven, CT, USA

Supplemental Methods.

Prespecified subgroups

To evaluate the consistency of the treatment effects across various subgroups, the estimate of the between-group treatment effect (with a nominal 95% CI) for the primary endpoints was estimated and plotted within each category of the following classification variables:

- Nodal status: Positive vs. Negative
- Tumor size: T1/T2 vs. T3/T4
- Choice of Carboplatin: Q3W vs. Weekly
- Tumor PD-L1 status
- Menopausal status: pre-menopausal vs. post-menopausal
- Age: <65 years vs. ≥65 years
- Geographic region: Europe/Israel/North America/Australia vs. Asia vs. Rest of World
- Ethnic origin: Hispanic vs. Non-Hispanic
- ECOG performance status: 0 vs. 1
- HER2 status: IHC 2+ (but FISH-) vs. IHC 0-1+
- LDH: >upper limit of normal (ULN) vs. ≤ULN

Supplemental Results

Overall survival

The proportional hazards assumption does not hold for overall survival at this interim analysis. The hazard ratio for overall survival based on the protocol-specified Cox regression model is 0.66 (95% CI, 0.50-0.87).

Figure S1. Trial Enrollment and Outcomes of the Patients at the Seventh Interim Analysis. Shown are data until the data-cutoff date of March 22, 2024. Patients did not have to complete all neoadjuvant therapy to undergo surgery. No patients remained on treatment at the seventh interim analysis. ITT, intention-to-treat.

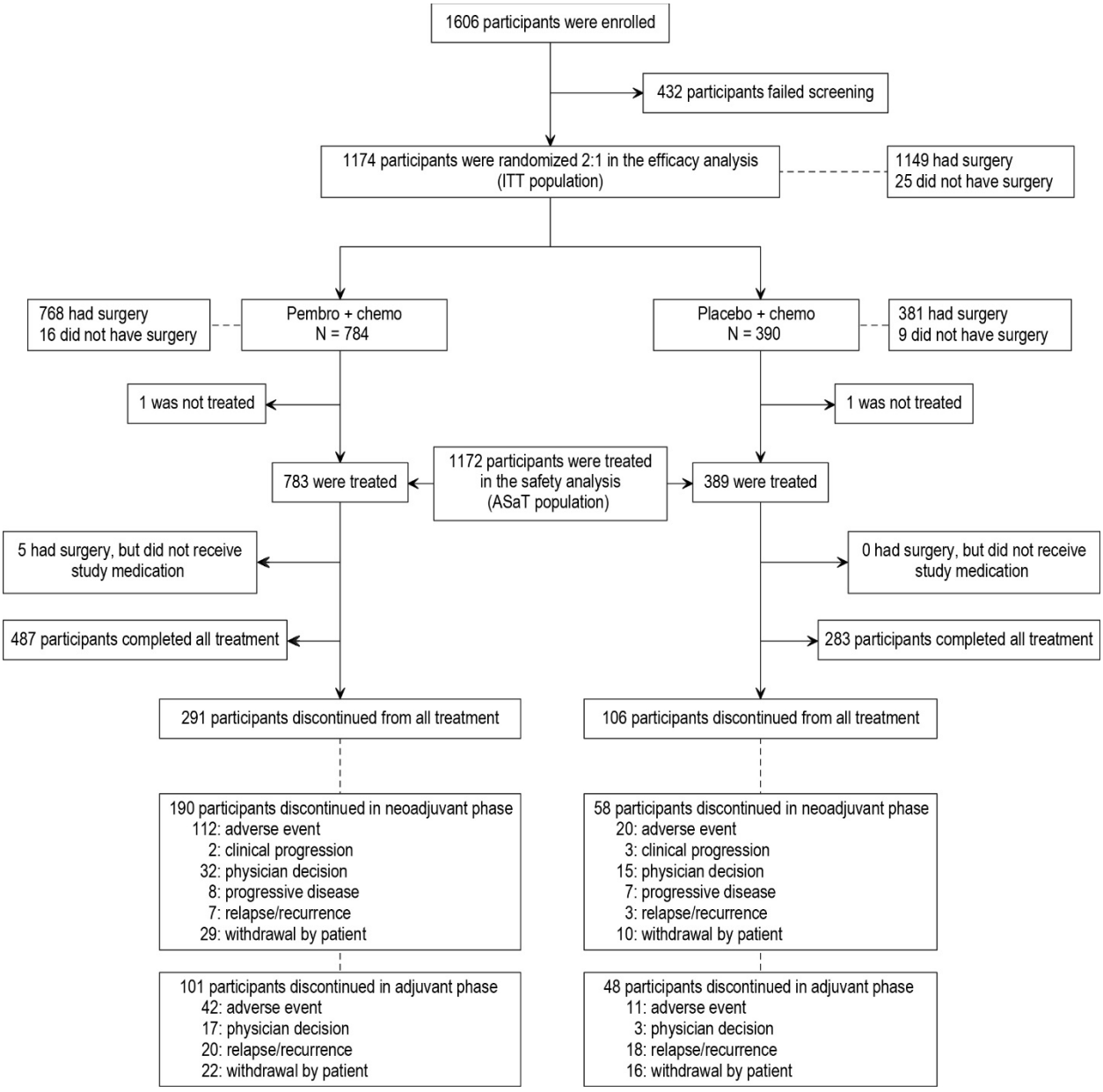


Fig. S2. Additional Subgroup Analyses of Overall Survival. Kaplan-Meier estimates of 5-year overall survival in additional prespecified subgroups is shown. The difference in 5-year overall survival and 95% confidence intervals for the difference are reported, assuming that the Kaplan-Meier survival rates at 5 years in both treatment groups follow the asymptomatic normal distribution independently. Eastern Cooperative Oncology Group (ECOG) performance-status (PS) scores range from 0 to 5, with higher scores indicating greater disability. Patients with missing values in a subgroup category variable are not included in the subgroup analysis.

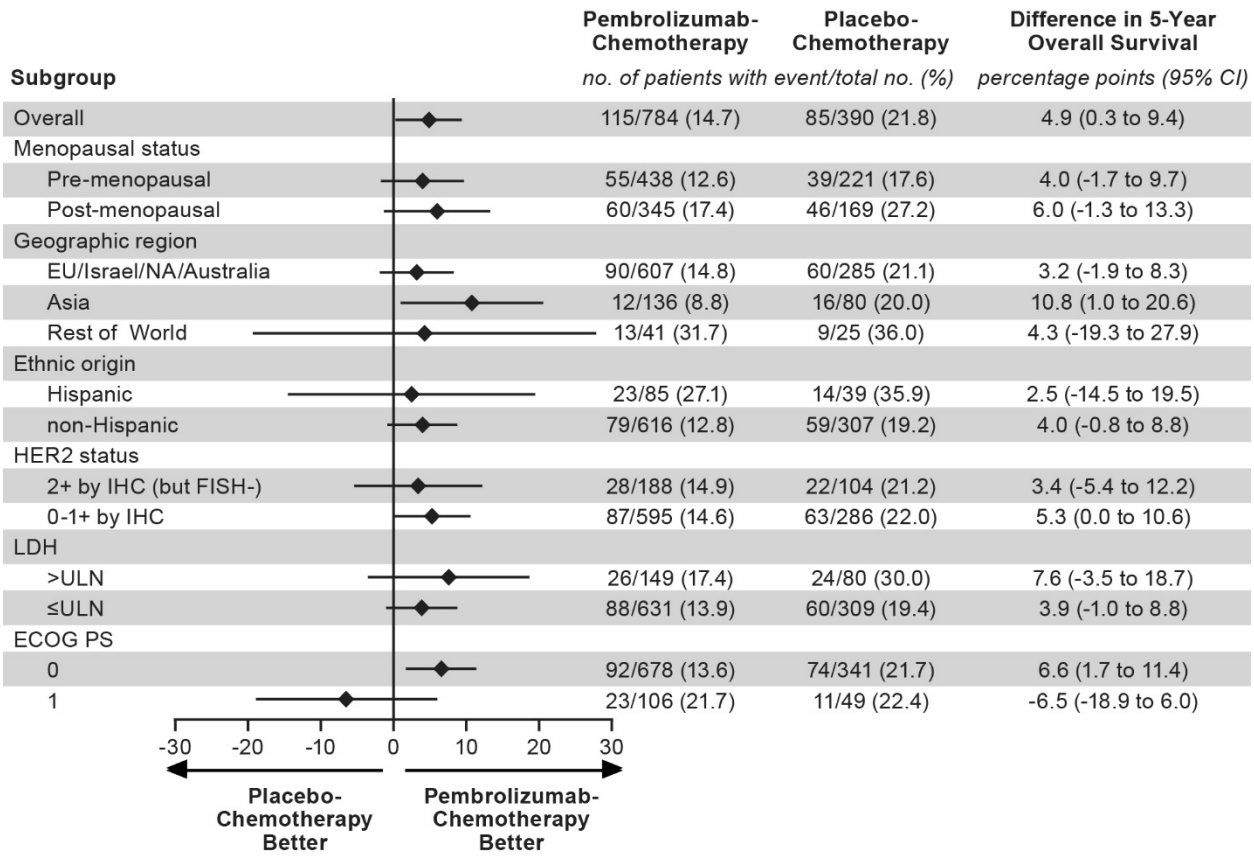


Fig. S3. Additional Subgroup Analyses of Event-Free Survival. An analysis of event-free survival in additional prespecified subgroups is shown. For the overall population, the analysis was based on the Cox regression model stratified according to nodal status (positive or negative), tumor size (T1 to T2 or T3 to T4), and frequency of carboplatin administration (once weekly or once every 3 weeks). For the other subgroups, the analysis was based on the unstratified Cox regression model. Eastern Cooperative Oncology Group (ECOG) performance-status (PS) scores range from 0 to 5, with higher scores indicating greater disability. Patients with missing values in a subgroup category variable are not included in the subgroup analysis. The proportional hazards assumption did not hold for the ECOG=1 subgroup; the 5-year rates of event-free survival were 75.2% in the pembrolizumab–chemotherapy group and 69.4% in the placebo–chemotherapy group (difference [95% CI] 5.8% [-9.5%, 21.2%]).

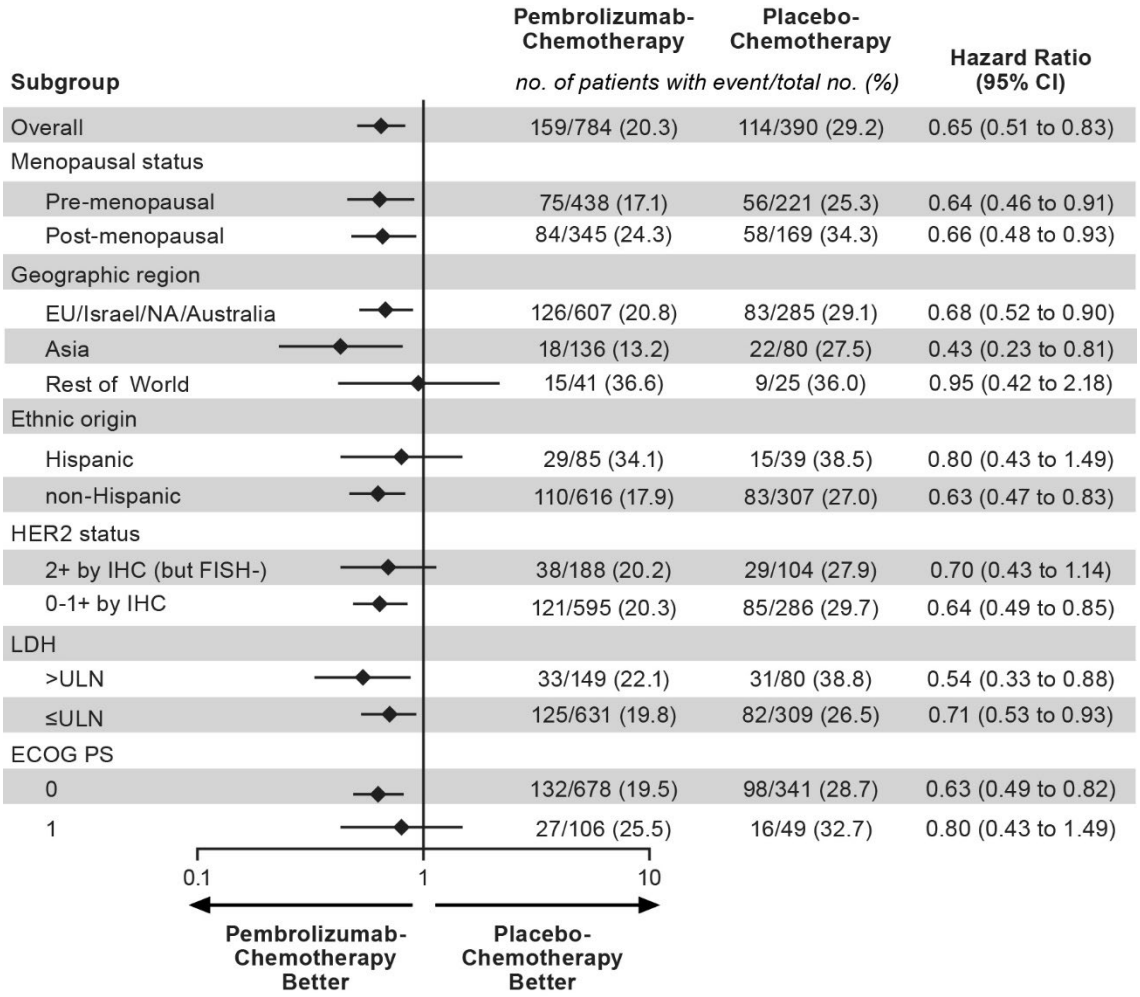


Fig. S4. Kaplan-Meier Estimates of Overall Survival by Pathological Complete Response (ypT0/Tis ypN0) According to Treatment Group in the Intention-to-Treat Population. Tick marks represent data censored at the last time the patient was known to be alive.

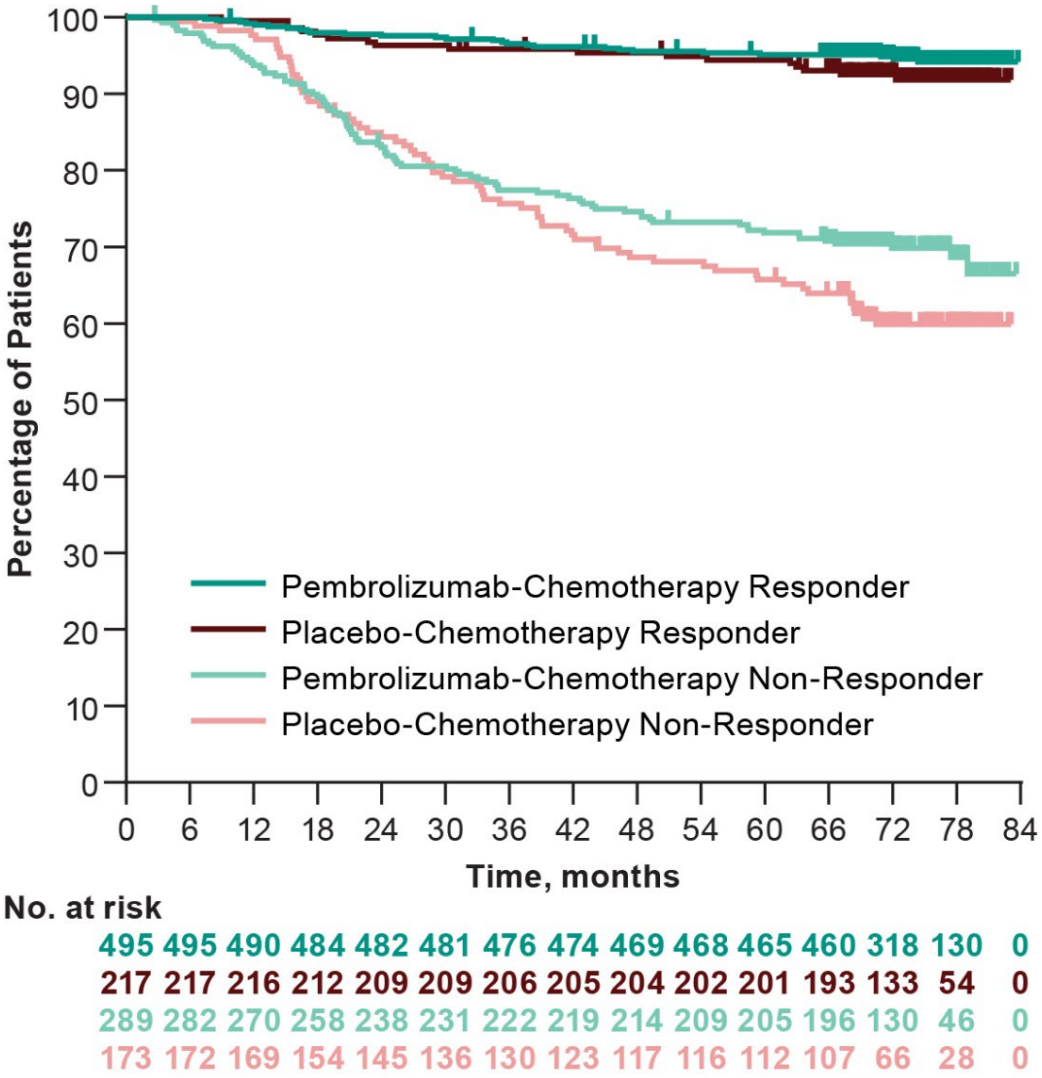


Table S1. Demographic and Disease Characteristics at Baseline.*

Characteristic	Pembrolizumab–Chemotherapy (N=784)	Placebo–Chemotherapy (N=390)
Age		
Median (range) — yr	49 (22-80)	48 (24-79)
<65 yr — no. (%)	700 (89.3)	342 (87.7)
Female sex — no. (%)	783 (99.9)	390 (100.0)
Geographic region — no. (%)		
Europe	388 (49.5)	180 (46.2)
Asia	166 (21.2)	91 (23.3)
North America	166 (21.2)	78 (20.0)
Australia	23 (2.9)	16 (4.1)
Rest of world	41 (5.2)	25 (6.4)
Race		
White	504 (64.3)	242 (62.1)
Asian	149 (19.0)	89 (22.8)
Missing†	65 (8.3)	31 (7.9)
Black or African American	38 (4.8)	15 (3.8)
American Indian or Alaska Native	14 (1.8)	7 (1.8)
Multiple	13 (1.7)	6 (1.5)
Native Hawaiian or Other Pacific Islander	1 (0.1)	0
Ethnicity		
Not Hispanic or Latino	616 (78.6)	307 (78.7)
Hispanic or Latino	85 (10.8)	39 (10.0)
Not Reported	46 (5.9)	28 (7.2)
Unknown	19 (2.4)	11 (2.8)
Missing†	18 (2.3)	5 (1.3)
Menopausal status — no. (%)		
Pre-menopausal	438 (55.9)	221 (56.7)
Post-menopausal	345 (44.0)	169 (43.3)
Missing‡	1 (0.1)	0 (0.0)
PD-L1 status§ — no. (%)		
CPS ≥1	656 (83.7)	317 (81.3)
CPS <1	128 (16.3)	69 (17.7)
Missing	0 (0.0)	4 (1.0)
ECOG performance status		
0	678 (86.5)	341 (87.4)
1	106 (13.5)	49 (12.6)
Lactase dehydrogenase — no. (%)		
≤ULN	631 (80.5)	309 (79.2)
>ULN	149 (19.0)	80 (20.5)
Missing	4 (0.5)	1 (0.3)
Choice of carboplatin — no. (%)		
Every 3 weeks	335 (42.7)	167 (42.8)
Weekly	449 (57.3)	223 (57.2)

Primary tumor classification— no. (%)		
T1/T2	580 (74.0)	290 (74.4)
T3/T4	204 (26.0)	100 (25.6)
Nodal involvement — no. (%)		
Positive	405 (51.7)	200 (51.3)
Negative	379 (48.3)	190 (48.7)
Overall disease stage — no. (%)		
I	0 (0.0)	1 (0.3)
II	590 (75.3)	291 (74.6)
III	194 (24.7)	98 (25.1)
HER2 status — no. (%)		
0-1 by IHC	595 (75.9)	286 (73.3)
2+ by IHC[‖]	188 (24.0)	104 (26.7)
Missing	1 (0.1)	0 (0.0)

*Based on data from the intention-to-treat population. Percentages may not total 100 because of rounding. PD-L1 denotes programmed death ligand 1; ECOG denotes Eastern Cooperative Oncology Group; HER2 denotes human epidermal growth factor receptor 2; IHC denotes immunohistochemistry.

[†]Information on race and ethnicity of trial participants is based on self-report and unavailable in some countries due to local laws and practices.

[‡]From one male patient.

[§]The PD-L1 combined positive score was defined as number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by total number of tumor cells × 100.

[‖]All tumors with HER2 expression 2+ by IHC were negative for HER2 amplification by in situ hybridization.

Table S2. Representativeness of Patient Population.

Disease under investigation	Triple-negative breast cancer (TNBC).
Special considerations related to	
Sex and gender	Breast cancer is the most diagnosed cancer among women, and the leading cause of cancer deaths globally[1], with TNBC accounting for 15% to 20% of all breast cancer subtypes.[2] TNBC in men is extremely rare.
Age	Compared with other forms of breast cancer, rates of TNBC are higher in young persons, with many patients being diagnosed before the age of 50 years.[3,4]
Race or ethnic group	TNBC affects Black persons disproportionately in the United States.[5] Several studies suggest a similar rate of among Hispanic and non-Hispanic White persons.[6] Compared with non-Hispanic White persons, Asian persons have significantly lower rates of premenopausal TNBC, but rates do not decline with increasing age.[7]
Geography	In 2022, female breast cancer was the second leading cause of global cancer incidence, with approximately 2.3 million new cases, comprising 11.6% of all cancer cases.[1] Of the global breast cancer burden, it has been estimated that ~170 000 are TNBC.[8] Significant heterogeneity was reported among different groups of Black persons by birthplace, with the highest prevalence among US-born and Western-African–born Black women, followed by Caribbean-born, and Eastern-African-born Black women.[9]
Other considerations	The incidence of breast cancer and mortality rates vary greatly within and between countries. Ethnic predisposition, family history, genetic associations, and modifiable risk factors are inextricably intertwined.
Overall representativeness of this trial	The patients in this trial demonstrated the expected sex and age distribution. The proportion of Black persons who underwent randomization was small (4.5%). Among the patients enrolled in North America, 15.2% were Black, a relatively larger proportion than the total population distribution of Black persons in the United States. Most patients were enrolled in Europe (48.4%) and North America (20.8%); no patient was enrolled in Africa. The information on race and ethnicity in this study is based on self-report and unavailable in some countries due to local laws and practices.

1. Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin.* 2024;74(3):229-263.
2. Diana A, Franzese E, Centonze S, et al. Triple-Negative Breast Cancers: Systematic Review of the Literature on Molecular and Clinical Features with a Focus on Treatment with Innovative Drugs. *Current Oncology Reports.* 2018;20(10):76.
3. Sajid MT, Ahmed M, Azhar M, et al. Age-related frequency of triple negative breast cancer in women. *Journal of the College of Physicians and Surgeons-Pakistan: JCPSP.* 2014;24(6):400-403.

4. Jitariu AA, Cimpean AM, Ribatti D, Raica M. Triple negative breast cancer: the kiss of death. *Oncotarget.* 2017;8(28):46652-46662.
5. Trivers KF, Lund MJ, Porter PL, et al. The epidemiology of triple-negative breast cancer, including race. *Cancer Causes Control.* 2009;20:1071–1082.
6. Kwan ML, Kushi LH, Weltzien E, et al. Epidemiology of breast cancer subtypes in two prospective cohort studies of breast cancer survivors. *Breast Cancer Res.* 2009;11:R31.
7. Lin C-H, Yap YS, Lee K-H, et al. Contrasting epidemiology and clinicopathology of female breast cancer in Asians vs the US population. *J Natl Cancer Inst*. 2019;111:1298–1306.
8. Foulkes WD, Smith IE, Reis-Filho JS. Triple-negative breast cancer. *N Engl J Med.* 2010;363:1938-1948.
9. Sung H, DeSantis CE, Fedewa SA, et al. Breast cancer subtypes among Eastern-African-born Black women and other Black women in the United States. *Cancer.* 2019;125:3401–3411.

Table S3. Summary of Drug Exposure in the Combined Phases.

Event	Pembrolizumab–Chemotherapy (N=783)	Placebo–Chemotherapy (N=389)
All drugs		
n	778	389
Median weeks (range)	57.9 (0.1 – 95.3)	59.1 (0.1 – 86.1)
Pembrolizumab (200 mg Q3W)		
n	778	0
Median weeks (range)	57.9 (0.1 – 95.3)	0
Median no. administrations (range)	17.0 (1.0. – 17.0)	0
Placebo (Q3W)		
n	0	389
Median weeks (range)	0	59.1 (0.1 – 86.1)
Median no. administrations (range)	0	17.0 (1.0 – 17.0)
Carboplatin (weekly)		
n	444	220
Median weeks (range)	11.1 (2.0 – 26.1)	11.1 (2.3 – 17.1)
Median no administrations (range)	12.0 (3.0 – 12.0)	12.0 (3.0 – 12.0)
Carboplatin (Q3W)		
n	334	167
Median weeks (range)	9.6 (0.1 – 16.4)	9.4 (3.0 – 15.1)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (2.0 – 4.0)
Paclitaxel (weekly)		
n	778	389
Median weeks (range)	11.3 (0.1 – 26.1)	11.3 (0.1 – 17.1)
Median no. administrations (range)	12.0 (1.0 – 13.0)	12.0 (1.0 – 12.0)
Doxorubicin (Q3W)		
n	488	247
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no. administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)
Epirubicin (Q3W)		
n	238	122

Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 12.0)
Median no administrations (range)	4.0 (1.0 – 5.0)	4.0 (1.0 – 4.0)
Cyclophosphamide (Q3W)		
n	726	369
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)